

III B.Tech. I Semester MODEL QUESTION PAPER

SIGNALS & SYSTEMS

For EEE

Time: 3 Hrs.

Max. Marks: 70 M

Answer Question No.1 compulsorily

Answer **ONE Question** from **EACH UNIT**

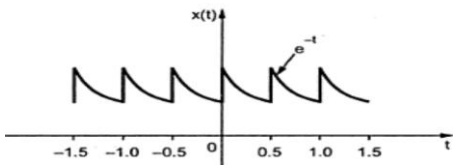
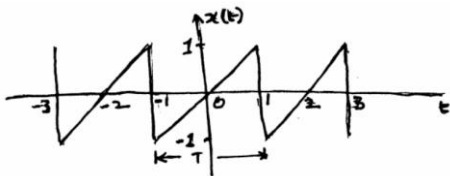
Assume suitable data if necessary

10 x 2 = 20 Marks

			CO	KL	M
1.	a).	Define unit step signal and unit impulse signal	1	2	2
	b).	Give the classification of systems	1	2	2
	c).	What are the properties of LTI systems?	2	2	2
	d).	Define singularity functions	2	2	2
	e).	Give the Fourier cosine series representation of a function.	3	2	2
	f).	What are the Dirichlet's conditions? State them.	3	2	2
	g).	Find the Fourier transform of $\delta(t - t_0)$	4	2	2
	h).	Find the Fourier Transform of $x[n] = 2^n u(n)$	4	2	2
	i).	State sampling theorem	5	2	2
	j).	Give the relation between Z-Transform and Fourier Transform	5	2	2

5 x 10 = 50 Marks

		UNIT-1			
2.	a).	Define signal power and signal energy. Check whether $x[n] = -0.5^n u(n)$ is a power signal or energy signal.	1	3	5
	b).	Test whether the following signal is periodic or not. If periodic Determine the fundamental period. $x(t) = 3\cos 10\pi t + \sin 20\pi t$	1	3	5
		OR			
3.	a).	Explain the system properties Linearity, Time-Invariance, Stability and Causality by giving examples.	1	3	5
	b).	Explain about the odd and even signals? Determine the even and odd components of the continuous-time signal $x(t) = e^{-t} u(t)$.	1	3	5
		UNIT-2			
4.	a).	Determine the Convolution between the two Signals $x(t) = e^{-3t} u(t)$ and $h(t) = tu(t)$.	2	3	5
	b).	Explain about the singularity functions and prove that $x(t) * \delta(t - t_0) = x(t - t_0)$	2	3	5

		OR			
5.	a).	Compute the convolution between $x[n] = \alpha^n u[n]$, $h[n] = \beta^n u[n]$.	2	3	5
	b).	Draw the block diagram representation of an LTI system in direct form-I realization for the system described by the difference equation $y(n)-14y(n-1)-18y(n-2) = x(n) + 3x(n-1) + 2x(n-2)$	2	3	5
		UNIT-3			
6.	a).	Prove that complex exponential signals are orthogonal functions.	3	4	5
	b).	Obtain the Exponential Fourier series for the periodic signal $x(t)$ shown below. 	3	4	5
		OR			
7.	a).	Determine the trigonometric Fourier series for the given signal shown below. 	3	4	5
	b).	Determine the best approximation of signal $x(t) = t$ in terms of $y(t) = \sin t$ over an interval $(-\pi \leq t \leq \pi)$.	3	4	5
		UNIT-4			
8.	a).	Derive the convolution property of the Continuous time Fourier Transform.	4	3	5
	b).	Find the Fourier Transform of $x(t) = e^{-2 t }$	4	3	5
		OR			
9.	a).	Find the inverse DTFT of $X(e^{j\omega}) = \frac{1}{(1 - ae^{-j\omega})^2}$ $ a < 1$	4	3	5
	b).	Obtain the step response of the system described by the following difference equation using DTFT. $y[n] - ay[n-1] = x[n]$ $ a < 1$	4	3	5
		UNIT-5			
10.	a).	Find the Z-transform of $x[n] = na^n u[n]$.	5	3	5
	b).	State and Explain about the Sampling theorem. What is "Aliasing effect" and how it can be avoided?	5	3	5
		OR			
11.	a).	Find the inverse Z-transform of	5	3	5

		$X(z) = \frac{z}{2Z^2 - 3Z + 1} \quad ROC: Z < \frac{1}{2}$			
	b).	A signal $x(t) = \cos 5\pi t + 0.5\cos 10\pi t$ is instantaneously sampled. Find the maximum interval of sampling from which the signal can be recovered.	5	3	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks



SRKR
ENGINEERING COLLEGE
AUTONOMOUS

Course Code: B23EE3102					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
III B.Tech. I Semester MODEL QUESTION PAPER					
TRANSMISSION & DISTRIBUTION SYSTEMS					
For EEE					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Discuss the any two advantages of high transmission voltage.	1	2	2
	b).	What is the need of primary transmission in an electric supply scheme?	1	2	2
	c).	Define the skin and proximity effects.	2	1	2
	d).	What are the limitations of nominal T and π methods?	2	2	2
	e).	What is meant by Ferranti effect?	3	1	2
	f).	What are the advantages of bundled conductor?	3	2	2
	g).	What is meant by radio Interference?	4	2	2
	h).	Write the properties of an insulator.	4	1	2
	i).	List the parts of underground cables.	5	1	2
	j).	Why do we use isolators on both sides of circuit breaker?	5	2	2
Fstd. 1980 AUTONOMOUS					
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Compare the volume of conductor material required for a d.c. 3-wire system and 3-phase, 3-wire system on the basis of equal maximum potential difference between one conductor and earth. Make suitable assumptions.	1	3	5
	b).	State and prove Kelvin's law for size of conductor for transmission. Discuss its Limitations.	1	3	5
		OR			
3.	a).	Compare AC transmission and DC transmission.	1	3	5
	b).	A 2 wire DC distributor ABCDEA in the form of a ring main is fed at point 'A' at 230V and is loaded as follows: 20A at B, 40A at C, 60A at D and 20A at E. The resistances of various sections (ground and return) are AB = 0.1 Ω , BC = 0.05 Ω , CD = 0.01 Ω , DE = 0.025 Ω and EA =0.075 Ω . Determine the point of minimum potential and current in each section of distributor.	1	3	5

		UNIT-2			
4.	a).	Derive an expression for the Inductance of a three-phase line having symmetrical spacing.	2	3	5
	b).	The three conductors of a 3-phase line are arranged at the corners of a triangle of sides 4, 5 and 6 meters. Calculate inductance per km of each conductor when conductors are regularly transposed. The diameter of each line conductor is 2cm.	2	3	5
		OR			
5.	a).	Derive an expression for the Capacitance of a single phase two wire line.	2	3	5
	b).	A 3-phase overhead transmission line has its conductors arranged at the corners of an equilateral triangle of 2 m side. Calculate the capacitance of each line conductor per km. Given that diameter of each conductor is 1.25cm.	2	3	5
		UNIT-3			
6.	a).	Derive the generalized circuit constants for short transmission line.	3	3	5
	b).	A 3phase, 50Hz, 150 km line has a resistance, inductive reactance and capacitive shunt admittance of 0.1Ω , 0.5Ω and $3 \times 10^{-6} S$ per km per phase. If the line delivers 50 MW at 110 kV and 0.8 p.f. lagging. Calculate the sending end voltage and current. Assume a nominal π circuit for the line.	3	3	5
		OR			
7.		Using rigorous method, derive expressions for sending end voltage and current for a long transmission line.	3	3	10
		UNIT-4			
8.	a).	Derive an Expression for the sag of transmission line conductor, suspended between two supports of the same height.	4	3	5
	b).	An overhead line has a span of 260 m; the weight of the line conductor is 0.68 kg per metre run. Calculate the maximum sag in the line. The maximum allowable tension in the line is 1550 kg.	4	3	5
		OR			
9.	a).	The self-capacitance of each unit in a string of three suspension insulators is C. The shunting capacitance of the connecting metal work of each insulator to earth is $0.15 C$ while for line it is $0.1 C$. Calculate (i) the voltage across each insulator as a percentage of the line voltage to earth and (ii) string efficiency.	4	3	5
	b).	Explain the following terms with reference to corona: (i) Critical Disruptive voltage (ii) Visual critical voltage (iii) Power loss due to corona.	4	3	5

		UNIT-5			
10.		Explain the following methods of cable grading: (i) Capacitance grading (ii) Intersheath grading	5	3	10
		OR			
11.	a).	Comparison between outdoor and indoor substation.	5	3	5
	b).	Discuss the different types of bus-bar arrangements.	5	3	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks



SRKR
ENGINEERING COLLEGE
AUTONOMOUS

Course Code: B23EE3103					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
III B.Tech. I Semester MODEL QUESTION PAPER					
POWER ELECTRONICS					
For EEE					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	What are the different methods to turn on the thyristor?	1	2	2
	b).	Power MOSFET is a voltage-controlled device. Why?	1	2	2
	c).	What are the advantages of freewheeling diodes in a controlled in a controlled rectifier?	2	2	2
	d).	What is meant by duty-cycle?	2	2	2
	e).	What are the applications of an inverter?	3	2	2
	f).	What is PWM? List the various PWM techniques.	3	2	2
	g).	Define cycloconverter and what are different types of cycloconverters?	4	2	2
	h).	What are the applications of an AC voltage controller?	4	2	2
	i).	Define modulation index.	5	2	2
	j).	What are the types of a multilevel inverter?	5	2	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Explain the dynamic characteristics of a thyristor during its turn-on and turn-off processes with a neat diagram	1	3	5
	b).	Describe the different modes of operation of a thyristor with the help of its static V-I characteristics.	1	3	5
		OR			
3.	a).	Demonstrate the switching characteristics of power MOSFET.	1	3	5
	b).	Explain the operation of Optocoupler based gate drive circuit.	1	3	5
		UNIT-2			
4.	a).	Explain the working of 1-phase full wave-controlled rectifier with RL load with relevant output waveforms and find average and RMS output voltage.	2	4	5
	b).	A single-phase full converter is supplied with 230V, 50Hz input is feeding a load $R=10\Omega$ in series with a large inductance that makes the	2	4	5

		load current ripple free. For firing angle of $\alpha = \pi/3$, calculate average output voltage.			
		OR			
5.	a).	With the help of neat circuit diagram and associated waveforms discuss the operation of a buck converter.	2	4	5
	b).	Illustrate various control strategies for varying duty cycle in choppers.	2	4	5
		UNIT-3			
6.	a).	Describe the bipolar and unipolar Pulse Width Modulation with neat waveforms.	3	3	5
	b).	Describe the operation of single-phase full bridge inverter with RL load and draw its related waveforms.	3	3	5
		OR			
7.		Explain for 3-phase inverter for 120° Conduction Mode of operation with waveforms.	3	3	10
		UNIT-4			
8.	a).	Explain and derive the output voltage equation of AC voltage controller with RL load and also draw necessary waveforms.	4	4	5
	b).	A single-phase full wave ac voltage controller feeds a load of $R=20\Omega$ with an input voltage of 230V, 50HZ. firing angle for both the thyristors is 45° . Determine (a) rms value of output voltage (b) power delivered to load and input power factor and (c) average input current.	4	4	5
		OR			
9.	a).	Draw the circuit of single phase to single phase step down cycloconverter and explain its operation with related waveforms	4	3	5
	b).	Analyze the operation of Step-Up Cyclo-converters with waveforms.	4	3	5
		UNIT-5			
10.	a).	Describe the features of Diode Clamped Multilevel Inverter (DCMLI) and mention its advantages.	5	3	5
	b).	Explain the principle of operation of Flying capacitor multilevel inverter (FCMLI).	5	3	5
		OR			
11.	a).	Explain the fundamental concept of a multilevel inverter and mention advantages and applications.	5	3	5
	b).	Explain the principle of operation of H-Bridge Multilevel Inverter (HB-MLI).	5	3	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23EE3104					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
III B.Tech. I Semester MODEL QUESTION PAPER					
SOLAR AND WIND ENERGY SYSTEMS					
For EEE					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Draw the i-v characteristics of solar cell.	1	2	2
	b).	Estimate the Average solar radiation.	1	2	2
	c).	Give the classification of Solar cells.	2	2	2
	d).	List the various specifications that need to be considered for a typical Solar module.	2	2	2
	e).	Distinguish between buck, boost and buck – boost converters.	3	2	2
	f).	Write the algorithm for MPPT by using hill climbing method.	3	2	2
	g).	Distinguish between lift force and drag force.	4	2	2
	h).	Write down the expression of condition for maximum wind power.	4	2	2
	i).	Classify the wind energy system configurations.	5	2	2
	j).	Distinguish between fixed speed WECS and variable speed WECS.	5	2	2
5 x 10 = 50 Marks					
		UNIT-1	CO	KL	M
2.	a).	Explain the working of the following radiation measurement instruments Pyrheliometer.	1	3	5
	b).	Sketch the spectrum of radiation from sun available at the top of the atmosphere, at the sea level and compare it with that of a black body	1	3	5
		OR			
3.	a).	Explain the working of the following radiation measurement instruments Pyranometer	1	3	5
	b).	Explain the term fill factor and its importance as a performance parameter for a solar cell.	1	3	5
		UNIT-2			
4.	a).	Discuss the ratings of a PV Module.	2	4	5
	b).	Explain in detail about the Solar cell mismatch in a series module and effect of shadowing on a Solar module.	2	4	5

		OR			
5.	a).	Explain the wattage of the module and I – V Equation of PV Modules with necessary equations.	2	4	5
	b).	Distinguish between Solar Cell, module and an Array.	2	4	5
		UNIT-3			
6.	a).	Explain briefly about a standalone connected PV system	3	3	5
	b).	Explain the algorithm for a MPPT by using perturb and Observe method	3	3	5
		OR			
7.	a).	Explain briefly about the operation of grid connected PV system.	3	3	5
	b).	Explain the operation of buck, boost converters by using neat circuit diagrams.	3	3	5
		UNIT-4			
8.	a).	Describe with a neat sketch the working of a wind energy system with main components?	4	3	5
	b).	Explain about the active stall control and passive stall control of aerodynamic power control.	4	3	5
		OR			
9.	a).	Compare between horizontal and vertical axis wind mills.	4	3	5
	b).	Draw the power characteristics of a wind turbine.	4	3	5
		UNIT-5			
10.	a).	Illustrate Variable Speed induction generator WECS.	5	3	5
	b).	Explain SCIG wind energy system with full capacity power converters.	5	3	5
		OR			
11.	a).	Explain Variable-Speed Synchronous Generator WECS Configuration with Full-Capacity Back-to-Back Power Converters	5	3	5
	b).	Illustrate Variable-Speed Synchronous Generator WECS Configuration with Distributed Converters for Multi winding Generators.	5	3	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23EE3105					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
III B.Tech. I Semester MODEL QUESTION PAPER					
ENERGY STORAGE SYSTEMS					
For EEE					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer ONE Question from EACH UNIT					
All questions carry equal marks					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	What is the need of energy storage?	1	2	2
	b).	Give the energy storage techniques.	1	2	2
	c).	What are flow batteries? Give one example.	2	2	2
	d).	State advantage of a Lithium-ion battery compared to a Lead-acid battery.	2	2	2
	e).	State the advantages of Flywheel Energy Storage (FES)	3	2	2
	f).	What is the difference between sensible heat storage and latent heat storage?	3	2	2
	g).	Mention characteristics that differentiate Super-capacitors from traditional capacitors.	4	2	2
	h).	What are the ways in which hydrogen can be stored.	4	2	2
	i).	Explain the Application of Energy storage in DER Systems	5	2	2
	j).	How do energy storage systems facilitate the integration of renewable energy into the grid.	5	2	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Explain the Applications and Technical benefits of energy storage systems	1	3	5
	b).	List the Financial Benefits of energy storage systems	1	3	5
		OR			
3.		Discuss the Techno economic characteristics of energy storage systems.	1	3	10
		UNIT-2			
4.		Explain Construction and Principle of operation of Lead-acid (Pb-acid) Battery and mention advantages & disadvantages.	2	3	10
		OR			

5.		Explain Construction and Principle of operation of Lithium-ion Battery and mention the Applications.	2	3	10
		UNIT-3			
6.	a).	Explain the operation of Pumped Hydroelectric Storage (PHS) system.	3	3	5
	b).	Explain the operation of Flywheel Energy Storage (FES) system.	3	3	5
		OR			
7.	a).	Explain the mechanism of Sensible Heat Storage (SHS) system.	3	3	5
	b).	Compare PHS, CAES and FES	3	3	5
		UNIT-4			
8.	a).	Explain the construction and operation of Super Capacitors	4	3	5
	b).	Explain the construction and operation of Superconducting Magnets	4	3	5
		OR			
9.		Explain Construction and Principle of operation of Hydrogen Fuel Cell	4	3	10
		UNIT-5			
10.	a).	Explain the Application of Energy storage for Utilities and Grid Operators.	5	3	5
	b).	Explain the Application of Energy storage for Smart Grid	5	3	5
		OR			
11.	a).	Explain the Application of Energy storage for SCADA System	5	3	5
	b).	Explain the Application of Energy storage for Consumer.	5	3	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23EE3106					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
III B.Tech. I Semester MODEL QUESTION PAPER					
DISTRIBUTED GENERATION AND MICRO-GRIDS					
For EEE					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Define Distributed Generation (DG) and explain how it differs from traditional central station generation.	1	1	2
	b).	What is a key advantage of DG in terms of enhancing grid reliability?	1	2	2
	c).	Name two primary types of solar power generation used in DG and explain a key operational difference between them.	2	1	2
	d).	List two common types of grid interconnection options available for renewable DG systems.	2	2	2
	e).	List two types of mechanical energy storage systems.	3	3	2
	f).	Give one advantage of capacitor storage over battery storage.	3	2	2
	g).	Define a microgrid and mention two key components.	4	2	2
	h).	Distinguish between AC and DC microgrids.	4	2	2
	i).	What is the significance of stability analysis in microgrids?	5	2	2
	j).	Mention two common power quality issues in microgrids	5	2	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Compare distributed generation and central station generation in terms of reliability, cost, and efficiency.	1	3	5
	b).	Explain the structure and working of traditional power systems.	1	3	5
		OR			
3.	a).	Discuss the cost implications of Transmission and Distribution in power systems.	1	3	5
	b).	Elaborate on various planning methods for DG systems and compare them.	1	3	5
		UNIT-2			
4.	a).	Explain the working and components of a Solar PV-based DG system.	2	3	5
	b).	Discuss the role of wind energy in renewable DG and its operational constraints.	2	3	5

		OR			
5.	a).	Describe the structure and working of a fuel cell-powered DG system	2	3	5
	b).	Compare and contrast different types of renewable DG systems	2	3	5
		UNIT-3			
6.	a).	Describe different types of energy storage systems and their roles in microgrids.	3	3	5
	b).	Explain the construction, working, and advantages of SMES.	3	3	5
		OR			
7.	a).	Discuss various types of mechanical energy storage technologies.	3	3	5
	b).	Compare different energy storage technologies based on efficiency, cost, and application.	3	3	5
		UNIT-4			
8.	a).	Explain the concept of a microgrid along with its structure and configuration.	4	3	5
	b).	Compare AC and DC microgrids in terms of control, applications, and technical challenges.	4	3	5
		OR			
9.	a).	Describe the role of power electronics in enabling the integration and control of distributed energy resources in microgrids.	4	3	5
	b).	Discuss the operational modes of a microgrid. How does control differ in grid-connected and islanded modes?	4	3	5
		UNIT-5			
10.	a).	Explain the major power quality problems in microgrids and suggest possible mitigation techniques.	5	3	5
	b).	Describe the methods used for stability analysis of microgrids under dynamic conditions.	5	3	5
		OR			
11.	a).	Discuss the economic considerations in planning and operating a microgrid, including capital and operational costs.	5	3	5
	b).	What are smart microgrids? Discuss their features and how they differ from traditional microgrids.	5	3	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23EE3201					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
III B.Tech. II Semester MODEL QUESTION PAPER					
MICROPROCESSORS AND MICROCONTROLLERS					
For EEE					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Draw the pin diagram of 8085 microprocessor.	1	3	2
	b).	Explain the function of flag register?	1	3	2
	c).	Explain the instruction cycle?	2	3	2
	d).	Compare Machine language and Assembly Language.	2	3	2
	e).	Name the modes of operation of 8255?	3	3	2
	f).	Write a short note on A/D converters.	3	3	2
	g).	Write a short note on program status register.	4	3	2
	h).	Classify the registers in ARM Processor.	4	3	2
	i).	Difference between Branch and Branch link.	5	3	2
	j).	List the factors that affect the instruction encoding size.	5	3	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Illustrate briefly about bus structure of 8085	1	3	5
	b).	Demonstrate the different addressing modes of 8085 microprocessor with suitable examples	1	3	5
		OR			
3.		Illustrate the functional block diagram of 8085 microprocessor with a neat sketch.	1	3	10
		UNIT-2			
4.	a).	Illustrate the stack and subroutine with an example.	2	3	5
	b).	With the help of a timing diagram demonstrate the operation of a MVI B, 65H	2	3	5
		OR			
5.		Illustrate each group of instruction set of 8085 microprocessor with examples.	2	3	10

		UNIT-3			
6.	a).	Illustrate the functional block diagram of interfacing IC 8251(USART) with neat sketch.	3	3	5
	b).	Illustrate the functional block diagram of interfacing IC 8255A PPI with neat sketch	3	3	5
		OR			
7.	a).	Illustrate the operating modes of interfacing IC 8253 Programmable interval timer	3	3	5
	b).	Illustrate the block diagram of interfacing IC 8279 Keyboard /Display Interface with neat sketch	3	3	5
		UNIT-4			
8.		Illustrate about the architecture of ARM Processor with a neat diagram	4	3	10
		OR			
9.	a).	Illustrate the three phases of pipelined execution of an instruction	4	3	5
	b).	Illustrate the bit band aliasing.	4	3	5
		UNIT-5			
10.	a).	Illustrate Assembler directives and how they are different from assembly instructions.	5	3	5
	b).	Illustrate how the barrel shifter improves the execution speed.	5	3	5
		OR			
11.	a).	Illustrate the use of MOV instruction to construct 32-bit constant.	5	3	5
	b).	Illustrate different memory addressing mode are supported by Cortex-M based processor?	5	3	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

III B.Tech. II Semester MODEL QUESTION PAPER

CONTROL SYSTEMS

For EEE

Time: 3 Hrs.

Max. Marks: 70 M

Answer Question No.1 compulsorily

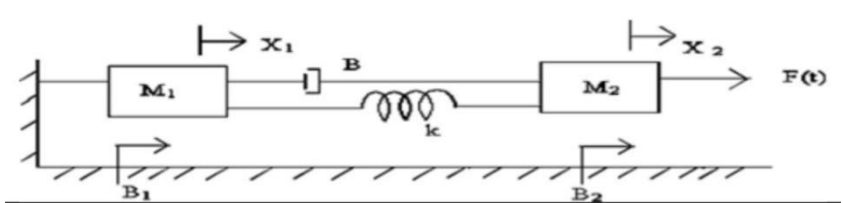
Answer **ONE Question** from **EACH UNIT**

Assume suitable data if necessary

10 x 2 = 20 Marks

			CO	KL	M
1.	a).	Define control system.	1	1	2
	b).	Mention any two block diagram reduction rules with diagram.	1	2	2
	c).	Define order and type of system with an example.	2	2	2
	d).	Define maximum peak overshoot and write formula.	2	2	2
	e).	Differentiate between Absolute stability and Relative stability.	3	2	2
	f).	Find centroid of a system whose characteristic equation is $1 + \frac{k}{s(s+1)(s+2)}$	3	3	2
	g).	Define Gain Margin and Phase Margin.	4	1	2
	h).	Define polar plot.	4	1	2
	i).	Write the advantages of state variable technique.	5	1	2
	j).	Write the properties of state transition matrix.	5	1	2

5 x 10 = 50 Marks

		UNIT-1			
2.	a).	Explain open-loop and closed loop control systems with examples.	1	3	4
	b).	Determine transfer function $X_1(S)/F(S)$ for the given mechanical translational system. 	1	3	6
		OR			
3.	a).	Find the transfer function C/R for the following system by using block diagram reduction	1	3	5

		technique			
	b).	Obtain the transfer function Y_6/Y_1 for the given SFG using Mason's gain rule.	1	3	5
		UNIT-2			
4.	a).	Derive the expression for unit-step response of a standard second-order under-damped system.	2	3	4
	b).	A unity feedback system is characterized by an open-loop transfer function $G(s) = K/S(S+10)$. Determine the value of gain K such that the system has a damping ratio of 0.5. With this value of K, find the settling time, percent overshoot and steady state error for a unit-ramp input.	2	3	6
		OR			
5.	a).	Discuss the effect of different controllers on the time response.	2	3	5
	b).	Determine the step, ramp, and parabolic error constants for the given unity feedback system. $G(s) = \frac{14(s+3)}{s(s+5)(s^2+2s+2)}$	2	3	5
		UNIT-3			
6.	a).	Explain Routh-Hurwitz stability criterion.	3	3	4
	b).	Determine the range of 'k' for which the characteristic equation $s^4 + 5s^3 + 5s^2 + 5s + k = 0$ is stable. Also find the frequency of sustained oscillations.	3	3	6
		OR			
7.		Sketch the root locus for the unity feedback system with following transfer function and determine the range of K for stability: $G(s) = \frac{K}{s(s^2+4s+13)}$	3	3	10

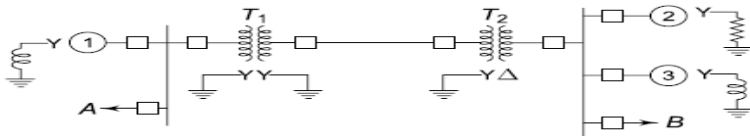
		UNIT-4			
8.		Sketch the bode plot for the following system and find gain & phase margin. $G(s) = \frac{50}{s(1+0.25s)(1+0.1s)}$	4	3	10
		OR			
9.	a).	The open loop transfer function of a unity feedback system is given by $G(S) = 1/S(S+2)$. Sketch the polar plot.	4	3	5
	b).	Explain the concept of Nyquist stability criterion.	4	3	5
		UNIT-5			
10.	a).	Determine the state model of the system whose transfer function is $\frac{Y(s)}{U(s)} = \frac{10(s+5)}{(s^3+6s^2+7s+8)}$	5	3	6
	b).	Derive transfer function from the state space model.	5	3	4
		OR			
11.	a).	Find the state transition matrix for the given system. $\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \end{bmatrix} u$	5	3	4
	b).	Test the controllability and observability condition for the following state space model $\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \\ \dot{x}_3(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & -2 & -3 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} u(t) \quad y = [3 \ 4 \ 1]x$	5	4	6

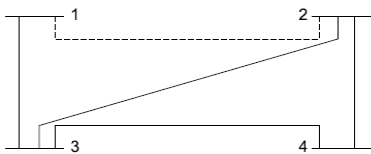
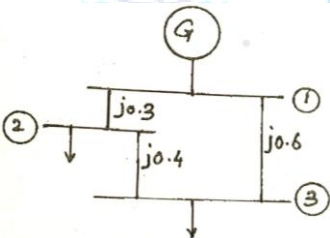
CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23EE3203					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
III B.Tech. II Semester MODEL QUESTION PAPER					
POWER SYSTEM ANALYSIS AND STABILITY					
For EEE					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
					10 x 2 = 20 Marks
			CO	KL	M
1.	a).	Find the base impedance for a 10 kV,5 MVA 3 –phase system.	1	3	2
	b).	A three phase transformer has a nameplate rating of 30 MVA, 230Y/69Y kV with a leakage -reactance of 10% and the transformer connection is wye-wye. Calculate the high voltage side impedance.	1	3	2
	c).	A power system has 200 buses of which 150 buses are load buses and others are generator buses. The size of the jacobian matrix is	2	3	2
	d).	How can the voltage of a particular bus be controlled	2	2	2
	e).	For what fault in the power system, the fault current is symmetrical.	3	2	2
	f).	Define positive and negative sequence components.	3	2	2
	g).	What is the value of (1+ a + a2) where a is complex operator.	4	2	2
	h).	What will be the value of IA , if Ia1 = 100 ∠ 30° and Ib2 = 20 ∠ 90°?	4	3	2
	i).	Name some of the disturbances which cause transient stability.	5	2	2
	j).	What is the range of ‘δ’ for stable operation?	5	2	2
					5 x 10 = 50 Marks
		UNIT-1			
2.	a).	What is single line diagram and explain with example how can reactance diagram be drawn from single line diagram.	1	3	5
	b).	Obtain the per unit impedance (reactance) diagram of the power system shown in Fig.  Generator No.1 : 30 MVA, 10.5 Kv, X'' = 1.6 ohms Generator No.2 : 15 MVA, 6.6 Kv, X'' = 1.2 ohms Generator No.3 : 25 MVA, 6.6 Kv, X'' = 0.56 ohms Transformer T1(3 phase): 15 MVA, 33/11 kV, X =15.2 ohms per phase on high tension side Transformer T2(3 phase): 15 MVA, 33/6.2 kV, X =16 ohms per phase on high tension side Transmission line : 20.5 ohms/phase	1	3	5

		OR																							
3.	a).	Define per unit quantity and obtain pu equivalent circuit of a single-phase transformer.	1	3	5																				
	b).	Obtain the PU equivalent circuit of three winding transformer.	1	3	5																				
		UNIT-2																							
4.	a).	Obtain Ybus of a simple four-bus system shown in fig. Table gives the line impedances. The shunt admittances are neglected. Find Y_{BUS} of the system assuming that the line shown dotted is connected.  <table><thead><tr><th>Line, bus to bus</th><th>R, pu</th><th>X, pu</th></tr></thead><tbody><tr><td>1-2</td><td>0.05</td><td>0.15</td></tr><tr><td>1-3</td><td>0.10</td><td>0.30</td></tr><tr><td>2-3</td><td>0.15</td><td>0.45</td></tr><tr><td>2-4</td><td>0.10</td><td>0.30</td></tr><tr><td>3-4</td><td>0.05</td><td>0.15</td></tr></tbody></table>	Line, bus to bus	R, pu	X, pu	1-2	0.05	0.15	1-3	0.10	0.30	2-3	0.15	0.45	2-4	0.10	0.30	3-4	0.05	0.15	2	4	5		
Line, bus to bus	R, pu	X, pu																							
1-2	0.05	0.15																							
1-3	0.10	0.30																							
2-3	0.15	0.45																							
2-4	0.10	0.30																							
3-4	0.05	0.15																							
	b).	Explain the Newton Raphson load flow algorithm to solve load flow problem.	2	3	5																				
		OR																							
5.	a).	For the power system shown below, find the voltages at the buses 2 & 3, at the end of 2 iterations by GS method.  <table><thead><tr><th>Bus</th><th>P(pu)</th><th>Q(pu)</th><th>V(pu)</th><th>Remarks</th></tr></thead><tbody><tr><td>1</td><td>-</td><td>-</td><td>1.03 +j0</td><td>slack</td></tr><tr><td>2</td><td>0.4</td><td>0.3</td><td>-</td><td>load</td></tr><tr><td>3</td><td>0.65</td><td>0.35</td><td>-</td><td>load</td></tr></tbody></table>	Bus	P(pu)	Q(pu)	V(pu)	Remarks	1	-	-	1.03 +j0	slack	2	0.4	0.3	-	load	3	0.65	0.35	-	load	2	4	5
Bus	P(pu)	Q(pu)	V(pu)	Remarks																					
1	-	-	1.03 +j0	slack																					
2	0.4	0.3	-	load																					
3	0.65	0.35	-	load																					
	b).	Compare advantages and disadvantages of gauss seidel and Newton Raphson methods.	2	3	5																				
		UNIT-3																							
6.	a).	Derive the expression for maximum momentary current in the short circuit of transmission line.	3	3	5																				
	b).	For the radial network shown in fig. a three-phase fault occurs at F. Determine the fault current and the line voltage at 11 kV bus under	3	4	5																				

		fault conditions.			
		OR			
7.	a).	Explain the behaviour of Synchronous machine during short circuit when it is operating on no load.	3	3	5
	b).	A power system consisting of two generators each of 10 MVA, 0.16 pu reactance connected to 11kv bus. A step up transformer 10 MVA, 0.1 pu reactance supplies power to 66 kv transmission line having impedance of $0.22+j0.22 \Omega / \text{km}$. the line is 40km long and at the other end, power is stepped down to 6.6 kv by 5MVA, 66/6.6 kv transformer of 0.08 pu reactance. A 3 ϕ fault occurs at the end of feeder having impedance of $0.44+j0.264 \Omega$ supplied from the load side bus of 6.6 kv. calculate the fault current at the fault point.	3	4	5
		UNIT-4			
8.	a).	A 400V, star connected, neutral grounded three phase generator is subjected to various types of faults. The fault currents for various types of faults are i) 120A for 3-phase fault, ii) 250A for line to ground fault, iii) 150A for line to line fault. If the resistances are neglected determine the three sequence impedances.	4	4	5
	b).	Derive the expression for fault current for a line to ground fault at the terminals of an unloaded alternator with fault impedance Z_f and draw the inter connection of sequence network.	4	3	5
		OR			
9.	a).	For a power system under abnormal conditions, the line currents are $I_R=10 \angle 30^\circ$, $I_Y=15 \angle -60^\circ$ and $I_B=18 \angle 154^\circ$ A. Find the symmetrical components of line currents.	4	3	5
	b).	A Synchronous generator 50 MVA, 13.8 KV has +ve, -ve and zero sequence reactance are j0.1, j0.1 and j0.08 p.u respectively. If LLG fault occurs at the terminals of the generator (neutral solidly grounded) find the fault current.	4	4	5
		UNIT-5			
10.	a).	Derive the swing equation of a single machine connected to infinite bus system.	5	3	5
	b).	Find the steady state power limit of a system consisting of a generator equivalent reactance 0.5 p.u connected to an infinite bus through a series reactance of 1.0 pu. The terminal voltage of the generator is held in 1.20 pu and the voltage of the infinite bus is 1.0 pu.	5	4	5
		OR			
11.	a).	Derive the expression for critical clearing angle and critical clearing	5	3	5

		time.			
	b).	A generator operating at 50Hz delivers 1.0 p.u. power to an infinite bus through a transmission circuit in which resistance is ignored. A fault takes place reducing the maximum power transferable to 0.4 p.u. whereas before the fault, this power was 1.8 p.u. and after the clearance of fault it is 1.3 p.u. By the use of equal area criterion, Determine the critical clearing angle.	5	4	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks



SRKR
ENGINEERING COLLEGE
AUTONOMOUS

Course Code: B23EE3204					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
III B.Tech. II Semester MODEL QUESTION PAPER					
POWER ELECTRONICS FOR RENEWABLE ENERGY					
For EEE					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Explain the two advantages of integrating renewables with Power electronics	1	2	2
	b).	Explain how power is generated from Solar?	1	2	2
	c).	Explain mismatch losses	2	2	2
	d).	Differentiate between string and array	2	2	2
	e).	Explain the role of gear box in wind energy system	3	2	2
	f).	Classify differnt types of wind generators	3	2	2
	g).	Write the application of fuel cell	4	2	2
	h).	Differentiate between battery and fuel cell	4	2	2
	i).	What is meant by grid synchronization?	5	2	2
	j).	Explain the role of PLL in grid synchronization?	5	2	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Classify the energy sources and explain the importance of renewable energy sources?	1	3	5
	b).	Explain the attributes of power electronics to Renewable energy systems.	1	3	5
		OR			
3.	a).	Describe the working principle of PEMFC with the help of diagram	1	3	5
	b).	Explain the interconnection of renewables with power electronics	1	3	5
		UNIT-2			
4.	a).	Draw and explain the operating region of P-V curve for Boost converter using basic equations?	2	3	5
	b).	Explain the advantages and disadvantages of single stage and two stage configurations for PV standalone systems	2	3	5
		OR			
5.	a).	Explain the PV array topologies with neat diagram and explain which	2	3	5

		topology reduces mismatch losses under most shading conditions with reason?			
	b).	PV module specification are $V_{oc}=25\text{ V}$ and $I_{SC}=2\text{ A}$. The load is 3-phase induction motor of rating 1 hp. If the power electronic configuration is single stage system and SPWM is used to generate the pulses for inverter. If N number of PV modules are provided design a PV array for the given load specification?	2	3	5
		UNIT-3			
6.	a).	Classify the merits and demerits of Type-III and Type-IV wind energy systems	3	3	5
	b).	Explain the turbine control methods for wind energy systems	3	3	5
		OR			
7.	a).	Explain the operation of Back-to-Back PWM VSI for Full Converter Turbines	3	3	5
	b).	Explain the principal of operation of PMSG?	3	3	5
		UNIT-4			
8.		Explain the use of Full bridge converter for a fuel cell-based power generation system	4	3	10
		OR			
9.	a).	Explain basic configurations of power conversion system (PCS) for Fuel cells?	4	2	5
	b).	Differentiate between isolated and non-isolated converters used in fuel cells?	4	3	5
		UNIT-5			
10.		Explain dq current control of voltage source converters for PV Grid integration	5	3	10
		OR			
11.	a).	Describe the difficulties encountered while integration renewable energy resources with the grid?	5	3	5
	b).	Explain the basic necessity of PLL in grid connected renewable systems?	5	3	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23EE3205					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
III B.Tech. II Semester MODEL QUESTION PAPER					
ELECTRICAL DISTRIBUTION SYSTEMS					
For EEE					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Write the relationship between maximum demand, connected load, and demand factor.	1	2	2
	b).	What is a load duration curve?	1	2	2
	c).	List two advantages of using a radial feeder.	2	2	2
	d).	State any two design considerations of a primary feeder.	2	2	2
	e).	Write two differences between DC 2-wire and DC 3-wire systems.	3	2	2
	f).	What are the causes of power loss in a distribution system?	3	2	2
	g).	Differentiate between Type K and Type T fuse links.	4	2	2
	h).	What is a line sectionalizer?	4	2	2
	i).	Mention one advantage of switched capacitors over fixed ones.	5	2	2
	j).	Define power factor and give its ideal value.	5	2	2
5 x 10 =50Marks					
		UNIT-1			
2.	a).	Apply the typical load behaviour of industrial and commercial loads to design a basic daily load curve. Explain the key features.	1	3	5
	b).	Illustrate the following terms with suitable examples: (i) Load factor, (ii) Loss factor, (iii) Contribution factor and (iv) Diversity factor.	1	3	5
		OR			
3.	a).	Derive the relation between Load factor and Loss factor	1	3	5
	b).	A distribution feeder consists of 5 residential customers. The peak load demands of these customers are 25 kW, 75 kW, 150 kW, 100 kW, and 50 kW. Total connected load of these customers is 1000 kW. The coincidence factor of these loads is 0.7. Determine i) Peak Demand ii) Demand Factor iii)Load Diversity iv) Diversity factor	1	3	5
		UNIT-2			
4.	a).	Derive the voltage drop in feeder lines with uniformly distributed load	2	4	5

	b).	Discuss about voltage drop constant 'K' for feeder with different types of loading	2	4	5
		OR			
5.	a).	Draw a line diagram of a radial-type primary feeder. Mention the factors that influence the selection of primary feeders.	2	4	5
	b).	Analyze the benefits of choosing an optimal substation location in terms of reliability.	2	4	5
		UNIT-3			
6.		Explain the significance of voltage drop in distribution systems. Apply the appropriate formula to compute voltage drop when the line parameters and load density of a uniformly loaded feeder are provided.	3	3	10
		OR			
7.		Calculate voltage drop and power loss in the case of non-three phase primary lines	3	3	10
		UNIT-4			
8.	a).	Demonstrate the main objectives of distribution system protection.	4	3	5
	b).	Illustrate the co-ordination procedure between fuse and a circuit breaker	4	3	5
		OR			
9.	a).	Draw a simple coordination diagram for a recloser–fuse setup in a rural distribution line and explain its operation.	4	3	5
	b).	Apply the concept of coordination of protective devices in a distribution system and explain the general coordination procedure	4	3	5
		UNIT-5			
10.		Determine the most suitable location for capacitor placement based on voltage profile and load.	5	3	10
		OR			
11.		Demonstrate how capacitors are used to improve the power factor	5	3	10

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23EE3206					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
III B.Tech. II Semester MODEL QUESTION PAPER					
COMPUTER ARCHITECTURE & ORGANIZATION					
For EEE					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Explain the role of buses in computer system.	1	3	2
	b).	How does the register differ from memory in a computer system.	1	3	2
	c).	What are instruction cycles in computer architecture?	2	2	2
	d).	Define the term interrupt.	2	2	2
	e).	What is the significance of microprogramming in control units?	3	3	2
	f).	What are the four basic type of operations that need to be supported by an instruction set.	3	3	2
	g).	Explain the memory hierarchy?	4	2	2
	h).	What type of memory technology is used in hard disks? Why?	4	2	2
	i).	What are the different methods used for handling the situation when multiple interrupts occur?	5	2	2
	j).	Write down different Peripheral devices	5	2	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Draw the schematic diagram and discuss the working of arithmetic logic shift unit.	1	3	5
	b).	What is a bus? What are different types of buses used in computers? Discuss their uses.	1	3	5
		OR			
3.	a).	Discuss about logic micro-operations.	1	3	5
	b).	What are micro-operations? Discuss arithmetic micro-operations.	1	3	5
		UNIT-2			
4.	a).	Discuss about Memory Reference Instructions.	2	2	5
	b).	Define and discuss about the type of registers.	2	2	5
		OR			
5.	a).	What are the basic differences among a branch instruction, a call	2	2	5

		subroutine instruction, and program interrupt?			
	b).	Discuss the effect of clock frequency on the speed of operation of computers.	2	2	5
		UNIT-3			
6.	a).	What are addressing modes? List and explain different addressing modes with suitable examples.	3	3	5
	b).	Explain the concept of stack organization in a computer system and its significance.	3	3	5
		OR			
7.		With a neat block diagram, explain the basic organization of a microprogrammed control unit.	3	3	10
		UNIT-4			
8.		Discuss about memory management hardware.	4	2	10
		OR			
9.	a).	Explain various mapping procedures of cache memory with an example.	4	2	5
	b).	What do you mean by virtual memory? Discuss how paging helps in implementing virtual memory.	4	2	5
		UNIT-5			
10.	a).	Explain about Input-output interface.	5	2	5
	b).	What is direct memory access? Explain its advantages.	5	2	5
		OR			
11.	a).	Discuss about priority interrupt.	5	2	5
	b).	Explain in detail various I/O modes of transfer.	5	2	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23EE3208					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
III B.Tech. II Semester MODEL QUESTION PAPER					
HIGH VOLTAGE ENGINEERING					
For EEE					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	What is meant by electric field stress?	1	2	2
	b).	Name two numerical methods used for electric field computation.	1	2	2
	c).	Why are gases used as insulating media in high-voltage systems?	2	2	2
	d).	State Paschen's law.	2	2	2
	e).	What is the function of a Marx circuit in high-voltage generation?	3	2	2
	f).	What is the role of a Tesla coil in high AC voltage generation?	3	2	2
	g).	How does a capacitive voltage divider work?	4	2	2
	h).	What is a Rogowski coil, and how does it measure high currents?	4	2	2
	i).	Define power frequency testing for insulators as per Indian standards.	5	2	2
	j).	How is DC testing conducted on isolators?	5	2	2
Estd. 1980 AUTONOMOUS					
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Discuss about the finite difference method for electric field computation.	1	3	5
	b).	Describe various methods implemented for protection against lightning over-voltages in an electrical power system.	1	3	5
		OR			
3.	a).	Explain how the electric stress can be estimated and controlled.	1	3	5
	b).	What are the sources of switching surges?	1	3	5
		UNIT-2			
4.	a).	Explain difference between photo-ionization and photo-electric emission?	2	3	5
	b).	Explain how treeing and tracking leads to breakdown in solid insulating materials.	2	3	5
		OR			
5.	a).	What is thermal breakdown in solid dielectrics, and how is it practically	2	3	5

		more significant than other mechanics?			
	b).	Explain the various mechanisms of breakdown phenomenon in commercial liquids	2	3	5
		UNIT-3			
6.	a).	Explain with diagrams different types of rectifier circuits to produce high D.C Voltages.	3	3	5
	b).	Describe a modified multi-stage Marx circuit for generation of impulse voltages.	3	3	5
		OR			
7.	a).	Describe with a neat sketch the working of a Van de Graaff generator. What are the factors that limit the maximum voltage obtained?	3	3	5
	b).	Derive the equivalent impedance expression for the cascade transformer.	3	3	5
		UNIT-4			
8.	a).	What are the different types of resistive shunts used for impulse current measurements? Discuss their characteristics and limitation	4	3	5
	b).	Explain the principle and construction of an Electrostatic Voltmeter for very high voltages. What are its merits and demerits for high voltage ac measurements?	4	3	5
		OR			
9.	a).	Describe generating voltmeter used for measuring high d.c voltages. How does it compare with a potential divider for measuring high dc currents	4	3	5
	b).	Explain the different methods of high A.C voltage measurements with their relative merits and demerits.	4	3	5
		UNIT-5			
10.	a).	Describe the impulse voltage testing procedure for circuit breakers as per IEC standards.	5	3	5
	b).	Explain the DC testing procedure for transformers and its relevance as per Indian standards.	5	3	5
		OR			
11.	a).	Discuss the significance of DC testing for cables and the procedure as per IEC standards.	5	3	5
	b).	Explain the concept of insulation coordination and its application in transformer testing.	5	3	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23EE3209					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
III B.Tech. II Semester MODEL QUESTION PAPER					
SPECIAL ELECTRICAL MACHINES					
For EEE					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	List two applications of stepper motors?	1	3	2
	b).	What are the three main types of stepper motors?	1	2	2
	c).	What type of Commutation is used in BLDC Motor?	2	2	2
	d).	What is the main difference between a conventional DC motor and a PMBLDC motor?	2	2	2
	e).	How does a PMSM differ from a conventional synchronous motor?	3	3	2
	f).	What are the different control techniques used for PMSM?	3	3	2
	g).	Why is rotor position measurement necessary in SRM control?	4	3	2
	h).	What are the main constructional features of an SRM?	4	2	2
	i).	Explain the working principle of a SYRM?	5	2	2
	j).	Write the torque equation of a SYRM?	5	2	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Explain in detail the construction and working of variable reluctance stepper motor.	1	3	5
	b).	Explain briefly closed loop control of stepper motor.	1	3	5
		OR			
3.	a).	Explain the principle of operation of permanent magnet stepper motor with torque Vs angle characteristics.	1	3	5
	b).	Enumerate the various applications of stepper motor.	1	3	5
		UNIT-2			
4.	a).	Explain the construction PMBLDC also compare conventional DC motor and PMBLDC motor.	2	3	5
	b).	Explain the speed-torque characteristics of PMBLDC motor.	2	3	5
		OR			
5.	a).	Explain in detail about power controllers used for PMBLDC motor.	2	3	5
	b).	Explain the closed loop control scheme of a PMBLDC motor drive	2	3	5

		with a suitable schematic diagram.			
		UNIT-3			
6.	a).	Explain the construction and working principle of operation of PMSM.	3	3	5
	b).	Derive the expression for power input and torque of a PMSM. Explain how its torque speed characteristics are obtained.	3	3	5
		OR			
7.	a).	Enumerate the applications of PMSM.	3	3	5
	b).	With necessary diagrams, discuss about various power controllers used for PMSM.	3	3	5
		UNIT-4			
8.	a).	Explain the principle of operation and the advantages of switched reluctance motor.	4	3	5
	b).	Describe with a neat circuit any two configuration of power converters used for the control of switched reluctance motor.	4	3	5
		OR			
9.	a).	Explain the torque-speed characteristics of switched reluctance motors.	4	3	5
	b).	Derive the expressions for voltage and torque of Switched Reluctance machine	4	3	5
		UNIT-5			
10.	a).	Illustrate with neat diagram, the construction, working principle of radial type synchronous reluctance motor	5	3	5
	b).	Discuss the main advantages and disadvantages of synchronous reluctance motor.	5	3	5
		OR			
11.	a).	Illustrate the performance of the synchronous reluctance motor with neat phasor diagram.	5	3	5
	b).	Discuss the various applications of synchronous reluctance motor.	5	3	5

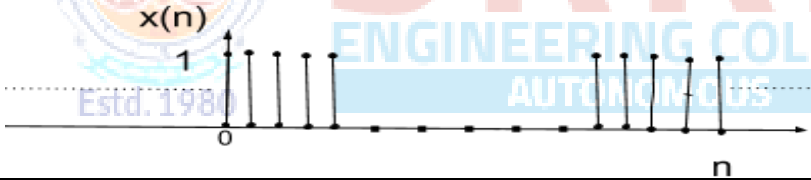
CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE : Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23EE3210					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					
III B.Tech. II Semester MODEL QUESTION PAPER					
DIGITAL SIGNAL PROCESSING					
For EEE					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	What are the advantages and limitations of Digital Signal Processing over Analog Signal Processing.	1	2	2
	b).	Given the difference equation: $y[n]=0.5y[n-1]+x[n]$ Find the system function $H(z)$ and check whether the system is stable.	1	2	2
	c).	State and explain any two properties of the Discrete-Time Fourier Transform (DTFT).	2	2	2
	d).	Compute the 4-point DFT of the sequence: $x[n]=\{1,2,3,4\}$.	2	2	2
	e).	Explain the difference between Decimation-in-Time (DIT) and Decimation-in-Frequency (DIF) FFT algorithms.	3	2	2
	f).	Using Radix-2 DIT-FFT, compute the 4-point FFT of the sequence $x[n]=\{1,0,-1,0\}$.	3	3	2
	g).	What is the bilinear transformation method in digital filter design? Mention its key feature.	4	2	2
	h).	Design a digital IIR low-pass filter using bilinear transformation for an analog filter with transfer function $H(s)=\frac{1}{s+1}$ Assume sampling time $T=1$ sec.	4	2	2
	i).	What is a linear-phase FIR filter? Mention its importance.	5	2	2
	j).	Using the Bartlett window , design a 3-tap FIR filter with the impulse response defined by: $h_d[n]=\frac{\sin(\omega_c(n-1))}{\pi(n-1)} \quad n=0,1,2$ where $\omega_c=\frac{\pi}{2}$ Compute the windowed impulse response $h[n]$ by multiplying $h_d[n]$ with the Bartlett window values.	5	3	2
5 x 10 = 50 Marks					
		UNIT-1			

2.	a).	Explain Sampling Theorem and Give the importance of Nyquist rate with respect to aliasing effect.	1	3	5
	b).	Consider the discrete time causal linear causal system defined by the difference equation $y(n) - \frac{3}{4}y(n-1) + \frac{1}{8}y(n-2) = x(n) + \frac{1}{3}x(n-1)$, Find system function H(z) and comment on stability? Implement system using Cascade connection by taking simple first order systems.	1	4	5
		OR			
3.	a).	Write Advantages and Applications of Digital Signal Processing	1	3	5
	b).	Obtain cascade realization of the following system $H(Z) = \frac{(1+\frac{3}{2}z^{-1}+\frac{1}{2}z^{-2})(1-\frac{3}{2}z^{-1}+z^{-2})}{(1+z^{-1}+\frac{1}{4}z^{-2})(1+\frac{1}{4}z^{-1}+\frac{1}{2}z^{-2})}$	1	4	5
		UNIT-2			
4.	a).	Find the frequency response of the system with unit sample response h(n)= 1, 0≤n≤N-1 and h(n)= 0, else where	2	3	5
	b).	Find the output of a filter whose impulse response is h(n)={1,1,1} and input signal x(n)={3,-1,0,1,3,2,0,1,2,1} using i) overlap-save method ii) overlap-add method	2	4	5
		OR			
5.	a).	Obtain Discrete Fourier series coefficient X(k) for the following periodic sequence x(n) 	2	3	5
	b).	Determine the output response y(n) if h(n)={1,1,1}; x(n)={1,2,3,1} by using i) linear convolution ii) circular convolution iii) linear convolution by using circular convolution.	2	4	5
		UNIT-3			
6.	a).	Explain radix-2 DIT FFT algorithm in detail by taking 8 points	3	3	5
	b).	Compute IDFT of the sequence X(K)={7,-0.707-j0.707,-j,0.707-j0.707,1,0.707+j0.707,j,-0.707+j0.707}	3	3	5
		OR			
7.	a).	Explain basic operations of DIT FFT and DIF FFT algorithms and draw their signal flow graphs.	3	3	5
	b).	Find the DFT of the sequence x(n)={1,2,2,1,1,2,2,1} using DIT FFT algorithm.	3	3	5
		UNIT-4			
8.	a).	Explain about IIR filter design using Bilinear transformation.	4	3	5
	b).	Design a digital Butterworth filter satisfying the constraints	4	4	5

		$0.707 \leq H(e^{jw}) \leq 1$ for $0 \leq w \leq \frac{\pi}{2}$ $ H(e^{jw}) \leq 0.2$ for $\frac{3\pi}{2} \leq w \leq \pi$ with $T=1$ sec using any IIR design method			
		OR			
9.	a).	Convert the single pole low pass filter with system function $H(z) = \frac{0.5(1+z^{-1})}{1-0.302z^{-2}}$ into bandpass filter with upper and lower cutoff frequencies ω_u and ω_l respectively. The lowpass filter has 3dB bandwidth $\omega_p = \frac{\pi}{6}$ and $\omega_u = \frac{3\pi}{4}$, $\omega_l = \frac{\pi}{4}$	4	4	5
	b).	Design a Third order Butterworth digital filter using impulse invariant technique. Assume sampling period $T=1$ sec.	4	4	5
		UNIT-5			
10.	a).	Compare IIR filter and FIR filter	5	3	5
	b).	Design an ideal low-pass filter with frequency response $ H(e^{j\omega}) = 1$ for $-\frac{\pi}{2} \leq \omega \leq \frac{\pi}{2}$ and $ H(e^{j\omega}) = 0$ for $\frac{\pi}{2} \leq \omega \leq \pi$ Find the values of $h(n)$ for $N=11$. Find $H(z)$. Plot the magnitude response	5	4	5
		OR			
11.	a).	Compare different window functions used in FIR filter design (Rectangular, Bartlett, Hamming, and Hanning).	5	3	5
	b).	Design a filter with $ H_d(e^{j\omega}) = e^{j3\omega}$ for $-\frac{\pi}{4} \leq \omega \leq \frac{\pi}{4}$ and $ H_d(e^{j\omega}) = 0$ for $\frac{\pi}{4} \leq \omega \leq \pi$ Using Hamming window with $N=7$.	5	4	5

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KL-KNOWLEDGE LEVEL

M-MARKS

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